

AquaSense – Firestore Schema Documentation

This document outlines the unified Firestore schema integrating Citizen Services, Supervisor Operations, IoT Monitoring, AI Prediction, and Hydration Tracking modules.

1. Collection: users

Document ID: uid (Firebase Auth)

Description: Stores profile data for all roles, including hydration settings.

Field Name	Data Type	Constraints	Description
name	String	Required	Full name of the user.
email	String	Required, Unique	User's email address.
phoneNumber	String	Unique	Verified 10-digit mobile number.
wardId	String	Required	Linked Ward ID (e.g., "Ward-A").
role	String	Required	Enum: 'citizen', 'supervisor', 'admin'.
profileImageUrl	String	Nullable	URL to profile picture.
hasActiveConnection	Boolean	Default: false	True if connection request is completed.
createdAt	Timestamp	Server Timestamp	Account creation time.
lastReadAnnouncements	Timestamp	Nullable	For notification badges.
hydrationGoal	Number	Default: 2000	Daily water intake goal in ml.
wakeUpTime	String	Nullable	"HH:MM" for hydration reminders.
bedTime	String	Nullable	"HH:MM" for hydration reminders.

2. Subcollection: users/{uid}/hydration_logs

Document ID: date (YYYY-MM-DD)

Description: Tracks daily hydration metrics.

Field Name	Data Type	Constraints	Description
currentIntake	Number	>= 0	Total ml consumed for this specific date.
dailyGoal	Number	Required	The goal that was set for this specific day.
history	Array	List of Maps	Log entries: [{time: Timestamp, amount: 200}].

3. Subcollection: users/{uid}/billingHistory

Document ID: Auto-generated bill ID

Description: Stores detailed billing cycles.

Field Name	Data Type	Constraints	Description
date	Timestamp	Required	Bill generation date.
amount	Number	Required	Bill amount (excluding fine).
reading	Number	Required	Meter reading value.
status	String	Enum: 'Due', 'Paid'	Payment status.
paymentId	String	Nullable	Razorpay ID or Cash Receipt ID.
generatedBy	String	Nullable	UID of Supervisor who created it.
fineAmount	Number	Default: 0	Late fee amount.

4. Collection: water_tanks

Document ID: wardId

Description: Real-time IoT tank telemetry.

Field Name	Data Type	Constraints	Description
tankName	String	Required	Display name (e.g., "Main Reservoir").
level	Number	0 - 100	Current water level percentage.
capacityLiters	Number	Required	Total tank capacity.
lastUpdated	Timestamp	Server Timestamp	Time of last update (Manual or IoT).
currentVolume	Number	Calculated	Real-time volume in Liters.
pressure	Number	Nullable	Pressure sensor reading (Bar/PSI).
flowRate	Number	Nullable	Flow sensor reading (L/min).
status	String	Enum	'Normal', 'Leak', 'Critical Low'.

5. Collection: sensor_logs

Document ID: Auto ID

Description: Time-series sensor dataset.

Field Name	Data Type	Constraints	Description
wardId	String	Required	Ward where the sensor is located.
type	String	Enum	'Ultrasonic', 'Flow', 'Pressure'.
value	Number	Required	Raw sensor reading value.
timestamp	Timestamp	Server Timestamp	Exact time of reading.

6. Collection: complaints

Document ID: Auto ID

Description: Citizen complaint workflow with AI classification.

Field Name	Data Type	Constraints	Description
userId	String	Required	UID of the citizen.
wardId	String	Required	Ward location of the issue.
description	String	Required	User's text description.
imageUrl	String	Required	Proof photo URL.
location	GeoPoint	Required	GPS Coordinates of issue.
status	String	Enum	'Submitted', 'In Progress', 'Resolved'.
createdAt	Timestamp	Server Timestamp	Submission time.
aiCategory	String	Nullable	AI Detected Label (e.g., "Pipe Burst").
aiConfidence	Number	0.0 - 1.0	Confidence score of the AI model.
resolutionImage	String	Nullable	Supervisor's proof of fix.

7. Collection: predictions

Document ID: Auto ID

Description: ML-driven water demand forecasting.

Field Name	Data Type	Constraints	Description
wardId	String	Required	Target Ward.
targetDate	Timestamp	Required	Future date for the prediction.
predictedDemand	Number	Required	Expected water usage (Liters).
modelUsed	String	String	Name of model (e.g., 'LinearRegression').
createdAt	Timestamp	Server Timestamp	When the prediction was run.

8. Collection: connection_requests

Document ID: Auto ID

Description: New connection onboarding system.

Field Name	Data Type	Constraints	Description
userId	String	Required	Applicant UID.
wardId	String	Required	Ward for connection.
address	String	Required	Full physical address.
residentialProofUrl	String	Required	Document URL (PDF/Image).
status	String	Enum	'Submitted', 'Approved', 'Rejected', 'Completed'.
appliedAt	Timestamp	Server Timestamp	Date of application.
supervisorProofUrl	String	Nullable	Image of installed meter (on completion).

9. Collection: announcements

Document ID: Auto ID

Description: Administrative broadcast engine.

Field Name	Data Type	Constraints	Description
title	String	Required	Headline.
message	String	Required	Body text.
wardId	String	Nullable	If null, it is Global. If set, only for that Ward.
isSupplyAlert	Boolean	Default: false	True for "Water Coming Soon" alerts.
supplyTime	Timestamp	Nullable	Scheduled time for water supply.
createdBy	String	Required	UID of Admin or Supervisor.

10. Collection: cash_collections

Document ID: Auto ID

Description: Cash settlement tracking.

Field Name	Data Type	Constraints	Description
supervisorId	String	Required	UID of collector.
amount	Number	Required	Total amount collected.
billIds	Array	List of Strings	IDs of bills covered by this collection.
status	String	Enum	'Pending', 'Settled'.
settlementTxnId	String	Nullable	Razorpay ID when transferred to Admin.

11. Collection: ward_pricing

Document ID: wardId

Description: Defines consumption pricing.

Field Name	Data Type	Constraints	Description
pricePerUnit	Number	Required	Cost per unit (kL) of water.

12. Collection: locations

Document ID: State

Description: Static location hierarchy.

Field Name	Data Type	Description
districts	Map	Key: District Name, Value: Array of Ward IDs.

13. Collection: phoneNumbers

Document ID: phoneNumber

Description: Ensures uniqueness in mobile numbers.

Field Name	Data Type	Description
uid	String	The UID associated with this number.